WNBA Fair Market Value Engine

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Erdös Institute
Summer 2026 Data Science Boot Camp

The WNBA Has a Valuation Problem

Context:

- The WNBA is experiencing unprecedented commercial growth. However, team salary caps remain strictly regulated.
- Front offices must optimize rosters with limited financial resources. They lack sophisticated statistical tools to help them choose players.

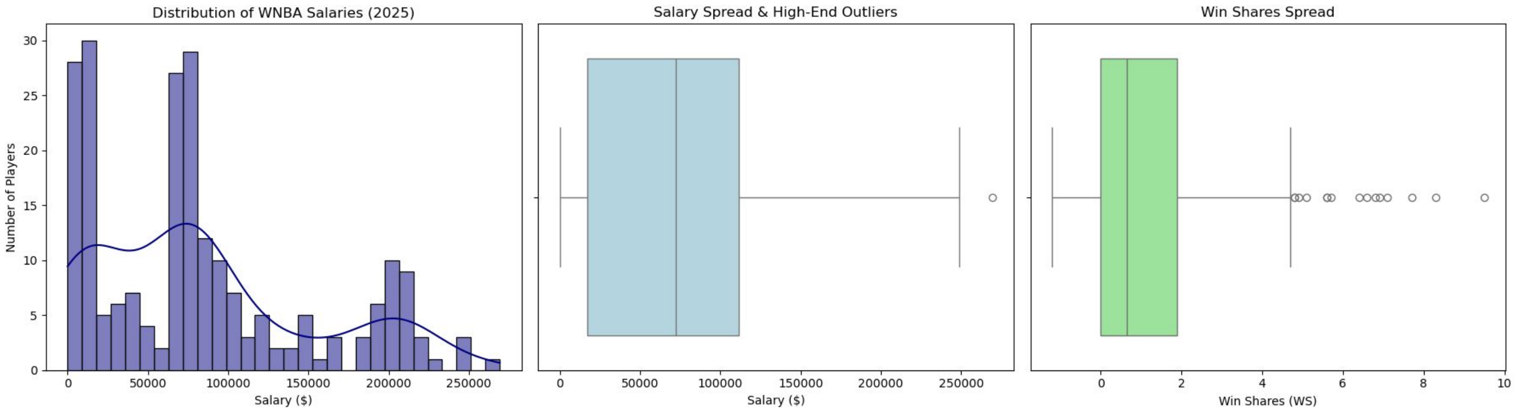
Approach:

- Develop a data-driven Fair Market Value (FMV) engine that predicts what a player should earn based on production.
- Compare predicted vs. actual contracts.

Goals:

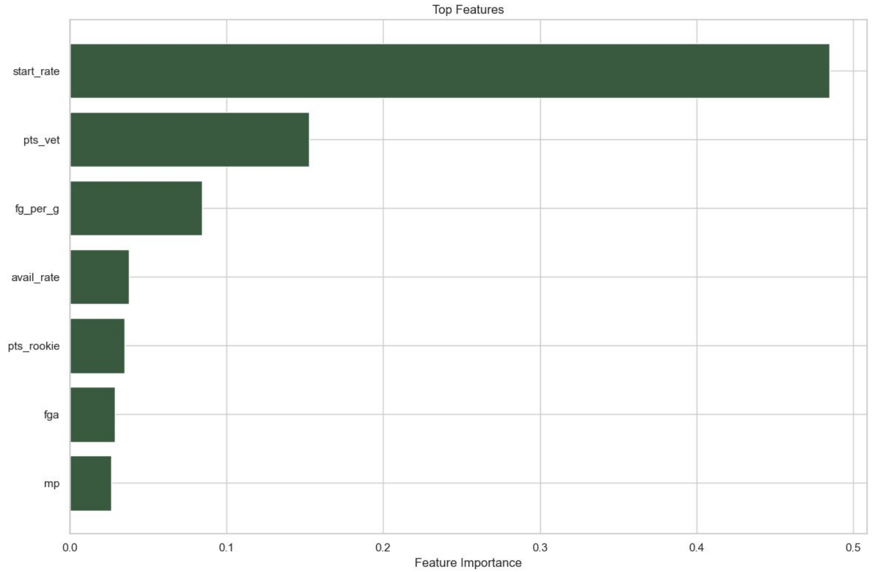
1. Identify market inefficiencies (under/over valued players).
2. Rank teams by spending efficiency relative to wins.

Five Seasons of Player and Salary Data



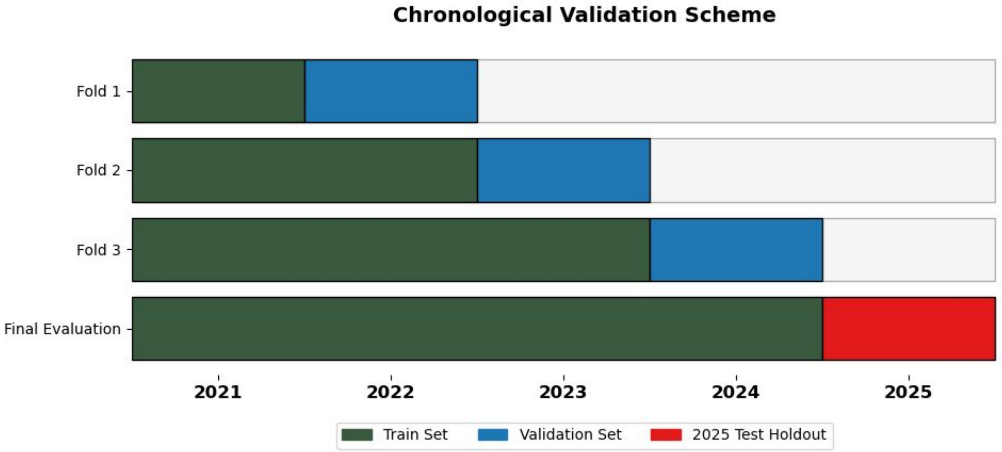
- Covers the 2021-2025 WNBA seasons with 250+ player records.
- Raw data has 150 features.
- Most features are player production statistics like points, assists, minutes played, and number of starts.
- Also includes salaries, contract types, and team context.
- Sources: HerHoopStats & Basketball Reference.

Isolating What Drives Player Valuation



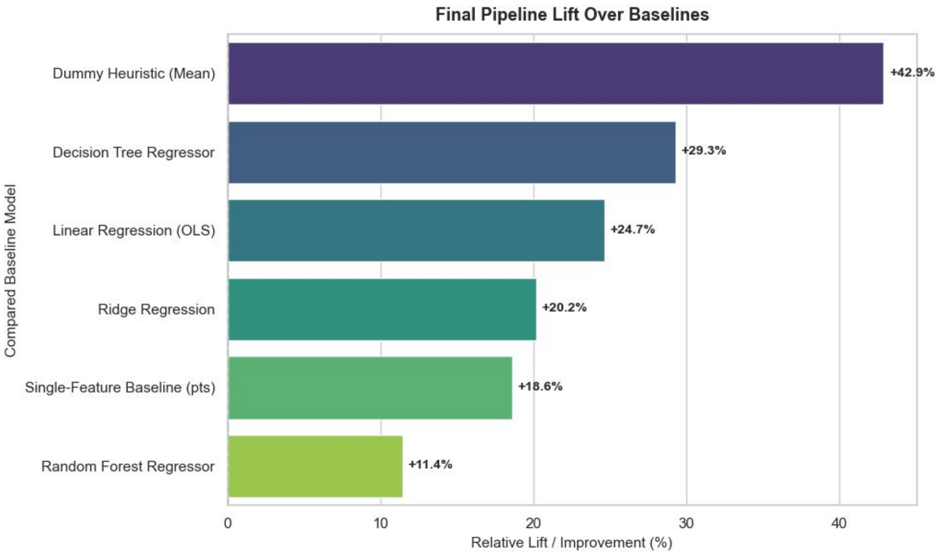
- Used SelectKBest, Lasso, and Random Forest to select 25 main features.
- All 3 methods suggested starting rate, points, and field goals per game as most salient player production features.
- Engineered features include availability rate, starting rate, and contract group indicators.

A Leakage-Proof Validation Approach



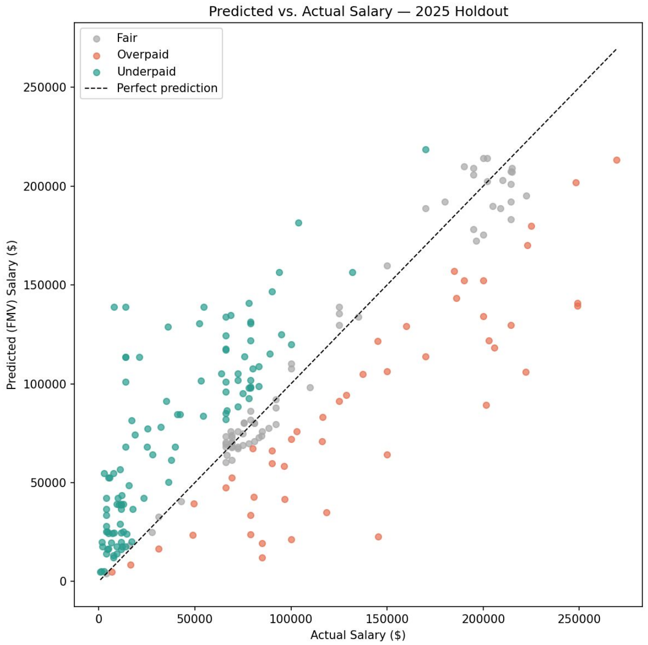
- Chronological split prevents temporal leakage by structuring training pools sequentially by year and leaving 2025 as a holdout.
- Individual players appear up to 5 times in our data, so we grouped cross-validation folds strictly by player identity.
- Raw data pairs salaries with concurrent performance, we frame features as lagged indicators so our model mimics actual decision-making.

FMV Engine Design and Performance



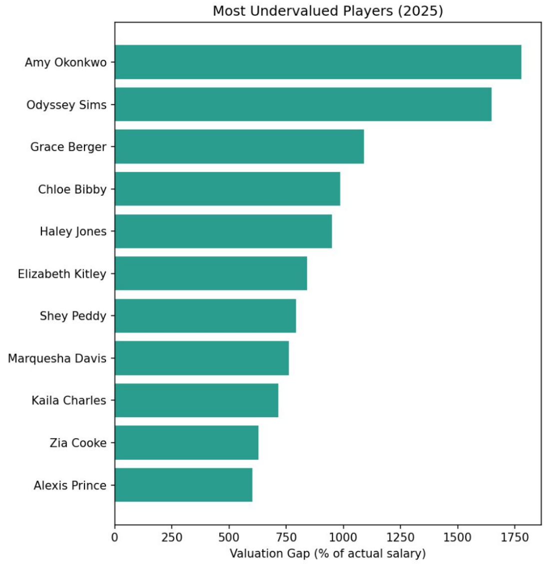
- Chose Root Mean Square Error (RMSE) as our Primary KPI.
- Final model is Random Forest with hyperparameters tuned using GridSearchCV (from scikit-learn). Final RMSE is $44k.
- Used bespoke baseline of a linear correlation between points and salary called Single-Feature to mimic naive predictions.

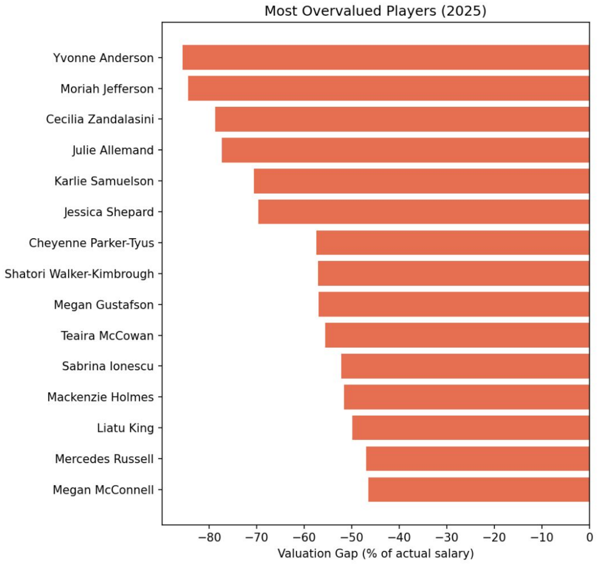
How Do We Classify Misvaluation?



- We map each player's actual 2025 contract salary directly against our engine's predicted Fair Market Value (FMV).
- Players are classified as overpaid or underpaid if their actual salary is more or less than their predicted salary ± 15%.

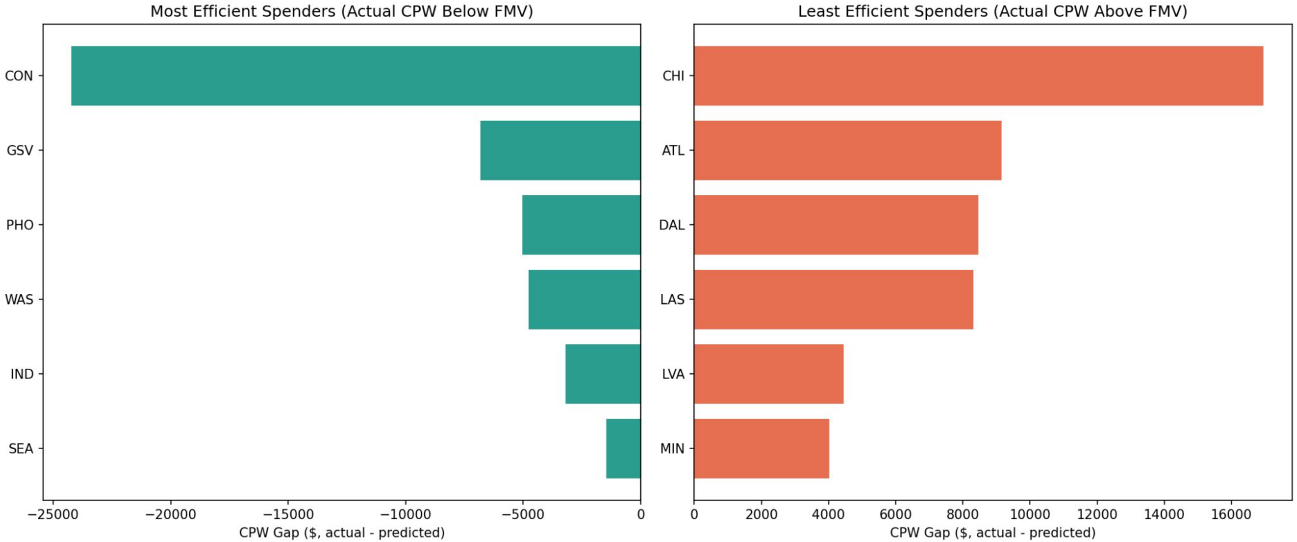
Who is the WNBA misvaluing?





- Players on hardship or rookie-scale contracts dominate the undervalued list because of league-wide restrictions on their salaries.
- Most overpaid players are veterans whose production has declined but whose salaries reflect past performance.

What Franchises Are Getting The Most From Their Payroll?



- CPW gap measures how much more (or less) a team spent per win than their roster's fair market value would predict.
- Results highlight different strategies: Seattle spent little but won sparingly, Las Vegas won big but slightly overpaid to do so.

What This Means in Practice

Conclusions

- Teams that find production-per-dollar value gain a roster edge.

Deployment

- GMs and players can use predicted salary as a negotiation tool.
- GMs and coaches can adjust recruitment strategies based on team-level efficiency scoring.
- Future versions could project multi-year salary trajectories.

Limitations

- Hardship contracts are structurally unlearnable.
- WNBA contract agreements change rapidly.
- Model does not capture factors like injury history or leadership.